

Cyber Security CTF Machine Development Report

INDEX PAGE

Sl. No.	Topic Name	Page No.
1	Title Page	1
2	Objective	2
3	Machine Specifications	2
4	Operating System Used	2
5	Network Configuration	3
6	User Accounts Created	3
7	Services Configured	4
8	Website Design Overview	4
9	Hidden Credentials Methodology	4
10	Steganography Implementation	5
11	Screenshots of Configuration	5-10
12	Conclusion	11

1. Title Page

Title: Cyber Security Capture The Flag (CTF) Machine Development

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Course: Cyber Security

Assignment: CTF Machine Development

Operating System: Kali Linux

Submission Date: 29/06/2026

2. Objective

The objective of this project is to design, configure, and deploy a Capture The Flag (CTF) machine using Kali Linux. The machine simulates a real-world penetration testing environment where participants perform reconnaissance, enumeration, exploitation, privilege escalation, and steganography analysis to retrieve hidden flags.

The project also demonstrates Linux system administration, web server configuration, secure user management, and cybersecurity concepts in a controlled environment.

3. Machine Specifications

Component	Specification
Virtualization Platform	VMware Workstation
Guest Operating System	Kali Linux
Processor	2 Virtual CPUs
RAM	4 GB
Storage	40 GB
Network Adapter	NAT / Host-Only
Hostname	Mokshit-MC

4. Operating System Used

The CTF machine was developed using **Kali Linux**, a Debian-based Linux distribution designed specifically for penetration testing and security assessments.

Features utilized include:

- Apache Web Server
- OpenSSH Server
- Linux User Management
- UFW Firewall
- Steghide
- Git
- Linux File Permissions

5. Network Configuration

The virtual machine was configured using VMware networking.

Configuration Details:

- Network Type: NAT (or Host-Only)
- IP Address: 192.168.152.128
- HTTP Port: 80
- SSH Port: 22

Machine discovery was performed using:

netdiscover

Open ports were verified using:

nmap -A 192.168.152.128

Only the following ports were accessible:

- Port 22 (SSH)
- Port 80 (HTTP)

Firewall configuration ensured all other ports remained closed.

6. User Accounts Created

Three standard Linux users were created.

Username

reva

User 2

User 3

Each user was assigned:

- Individual password
- Home directory
- Bash shell

Passwords were intentionally hidden within the website and were not disclosed directly.

7. Services Configured

The following services were installed and configured.

Apache Web Server

- Runs on Port 80
 - Hosts the CTF challenge website
 - Displays the personal profile
-

8. Website Design Overview

Website sections include:

- Index.html
- Robots.html
- Script.js
- Secret.html
- Style.css

The website also contains multiple hidden clues that guide participants toward discovering usernames, passwords, and hidden resources.

9. Hidden Credentials Methodology

Instead of displaying credentials openly, multiple information hiding techniques were used.

Methods include:

- HTML comments
- Hidden webpages
- JavaScript comments
- CSS comments
- robots.txt
- Base64 encoded strings
- Hidden directories
- Backup files

Participants must inspect the website source code and enumerate hidden resources to recover valid login credentials.

10. Steganography Implementation

An image was placed on the Desktop of the virtual machine.

A PDF version of the resume was embedded inside the image using Steghide.

Command used:

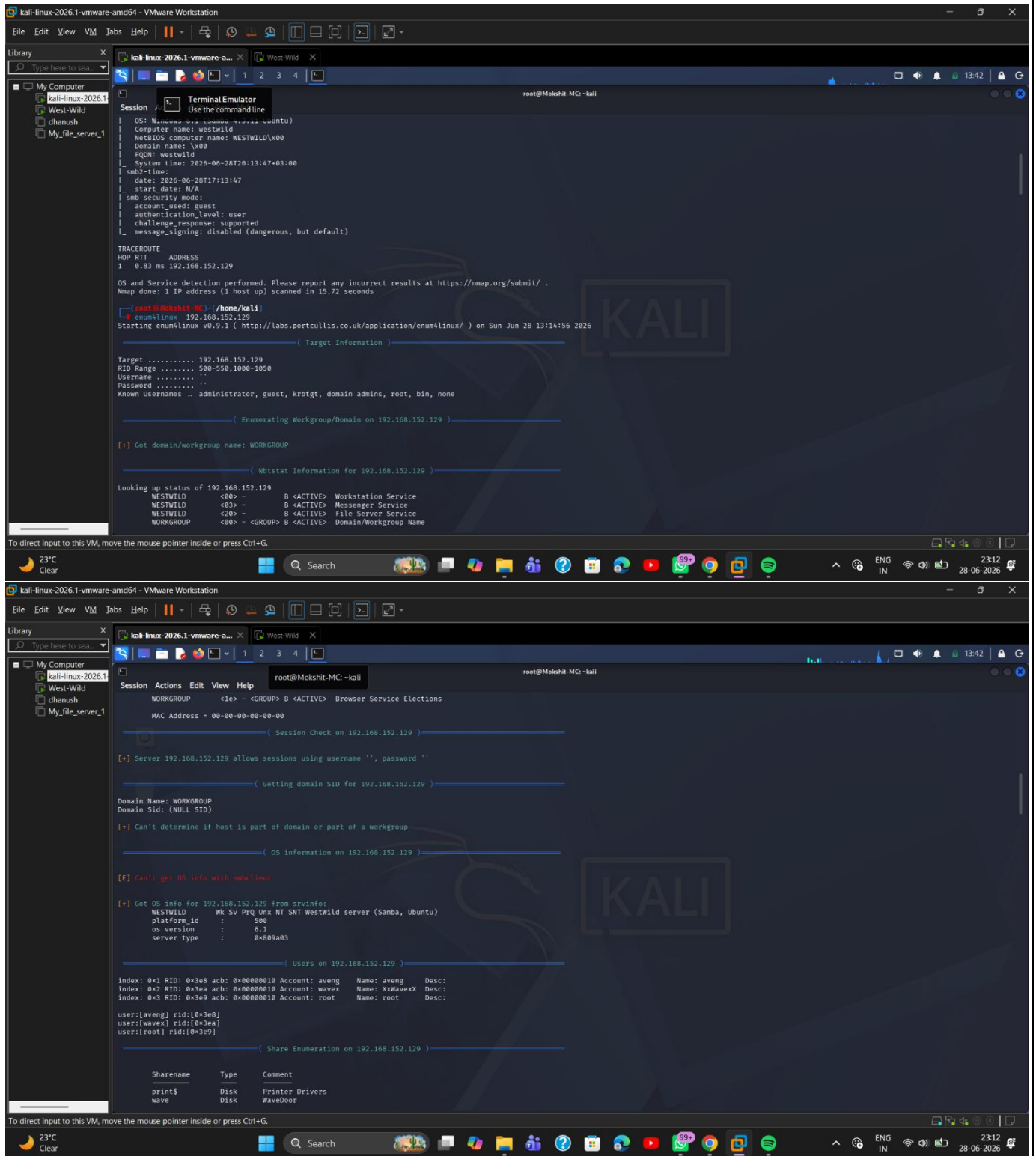
```
steghide embed -cf profile.jpg -ef resume.pdf
```

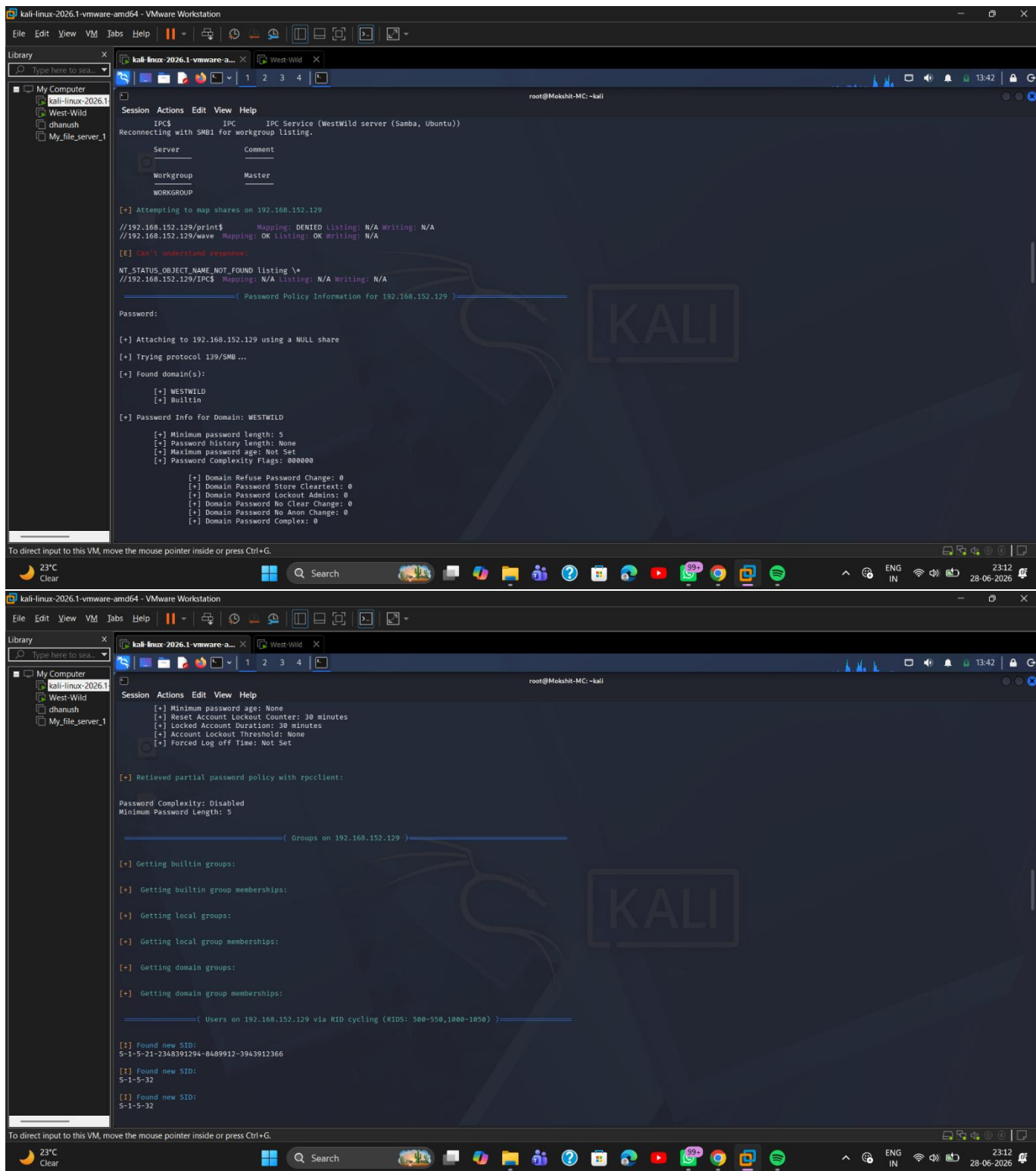
Participants are required to:

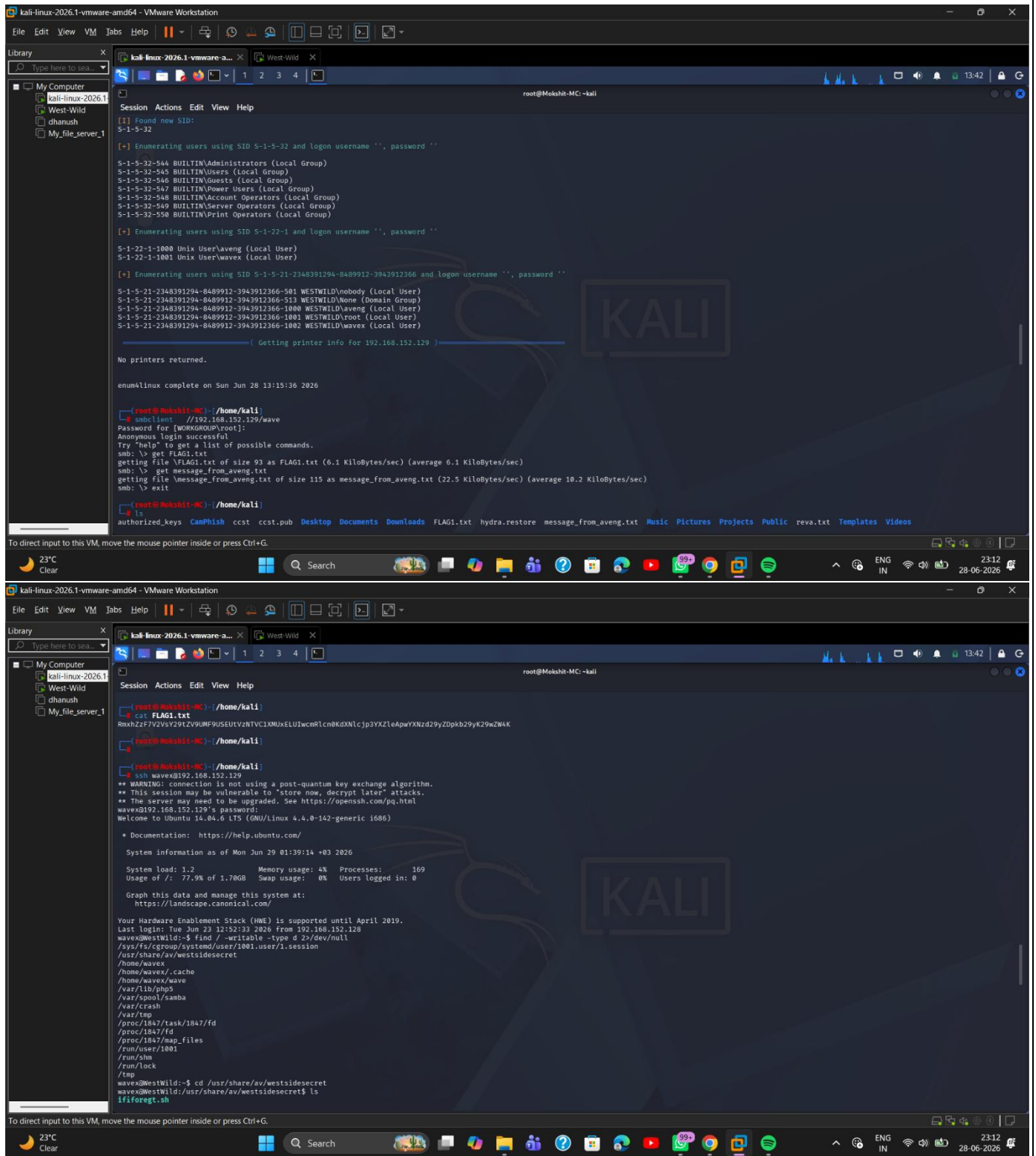
- Identify the image
- Discover the extraction password
- Extract the hidden resume

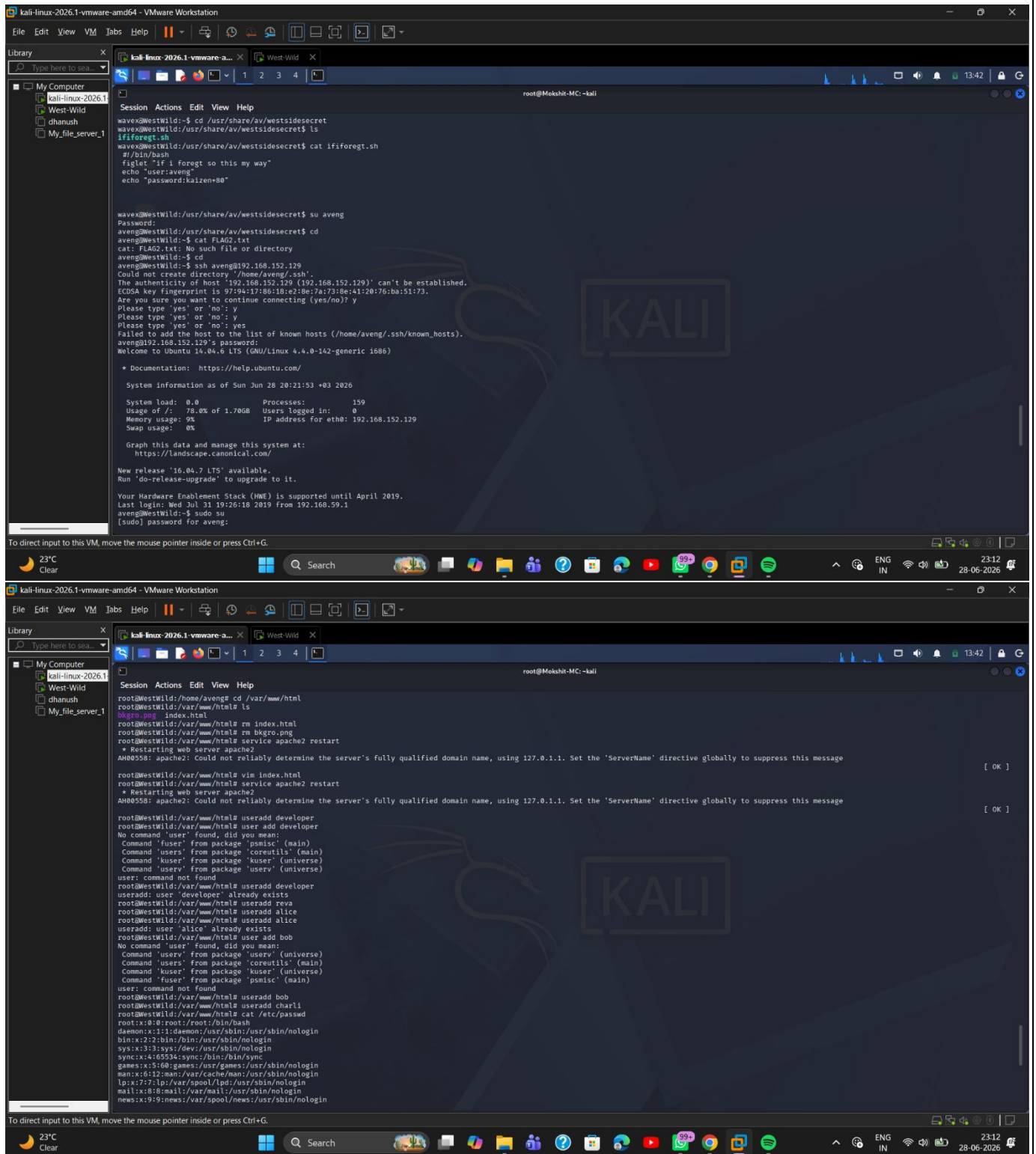
11. Screenshots of Configuration











kali-linux-2026.1-vmware-amd64 - VMware Workstation

File Edit View VM Tabs Help

Library

My Computer

kali-linux-2026.1

West-Wild

dhanush

My_file_server_1

Session Actions Edit View Help

```
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mail List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/var/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
libuid:x:100:100::/var/lib/libuid:/bin/false
syslog:x:101:104::/home/syslog:/bin/false
mysql:x:102:106:MySQL Server,/,/nonexistent:/bin/false
messagebus:x:103:107::/var/run/dbus:/bin/false
landscape:x:104:111::/var/lib/landscape:/bin/false
sshd:x:105:65534::/var/run/sshd:/usr/sbin/nologin
aveng:x:1000:1000:aveng,.,./home/aveng:/bin/bash
wawex:x:1001:1001:XWawex,.,./home/wawex:/bin/bash
developer:x:1002:1002::/home/developer:
reva:x:1003:1003::/home/reva:
alice:x:1004:1004::/home/alice:
bob:x:1005:1005::/home/bob:
charli:x:1006:1006::/home/charli:
root@WestWild:/var/www/html# passwd root
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
root@WestWild:/var/www/html# exit
exit
aveng@WestWild:~$ exit
logout
Connection to 192.168.152.129 closed.
aveng@WestWild:~$ exit
exit
wawex@WestWild:/usr/share/av/westsideseat$ exit
logout
Connection to 192.168.152.129 closed.

root@Mekshil-MC: ~ -kali
ssh reva@192.168.152.129
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
reva@192.168.152.129's password:
Permission denied, please try again.
reva@192.168.152.129's password:
Permission denied, please try again.
reva@192.168.152.129's password:
reva@192.168.152.129: Permission denied (publickey,password).
```

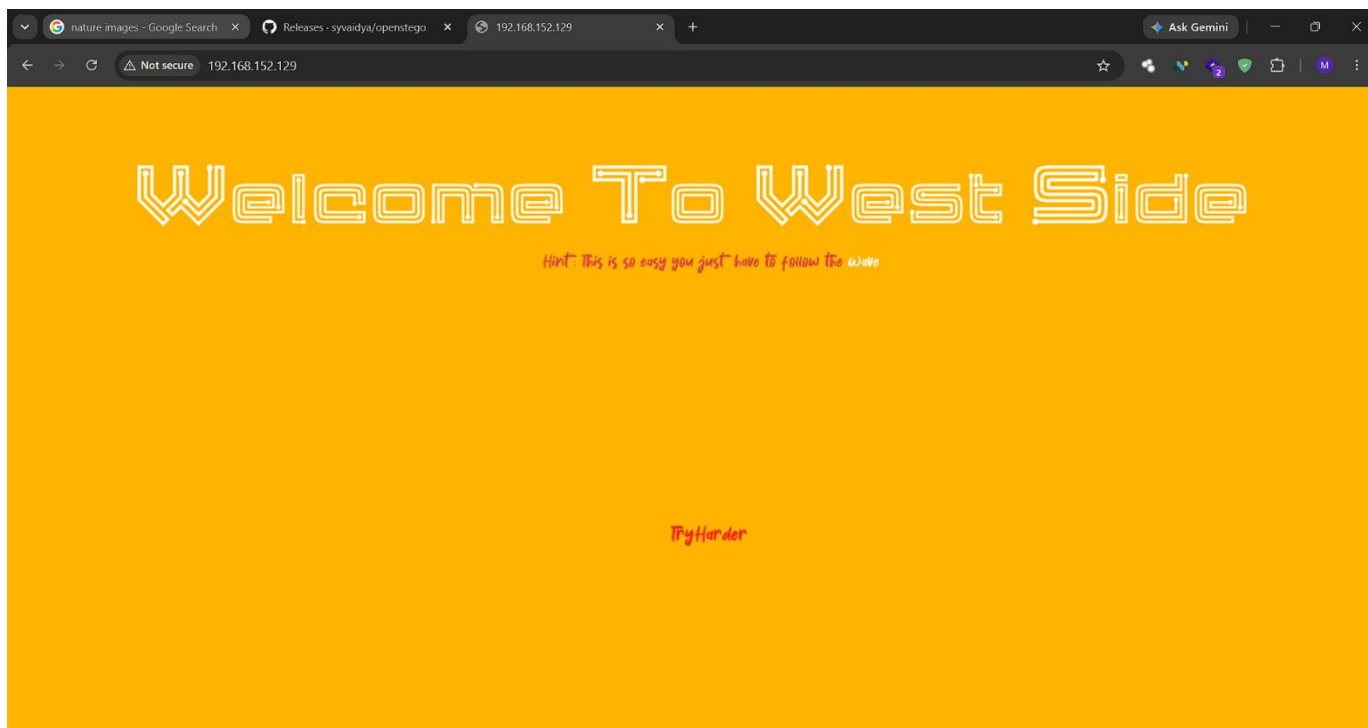
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

23°C Clear

Search

ENG IN

23:13 28-06-2026



12. Conclusion

This project successfully demonstrates the design and implementation of a beginner-to-intermediate Capture The Flag (CTF) machine using Kali Linux. The environment integrates Linux administration, secure service configuration, web application development, user management, steganography, and controlled vulnerability design into a single learning platform.

Participants gain hands-on experience in reconnaissance, enumeration, credential discovery, SSH access, privilege escalation, and digital forensics. The project reflects common penetration testing techniques used in real-world security assessments while providing a safe environment for learning and practice.

Overall, the CTF machine fulfills the assignment objectives and serves as an effective educational platform for developing practical cybersecurity skills.